

Your Instructor — Imran Ahmad, PhD

Google Cloud Authorized Trainer · Google Certified Professional – Data Engineer · AWS Authorized Instructor (AAI) · Microsoft Certified Trainer (MCT) · Author, Packt Publishing
— [linkedin.com/in/cloudanum](https://www.linkedin.com/in/cloudanum) · github.com/cloudanum · Ottawa, ON, Canada. 20+ years designing, building, and teaching large-scale data systems — across Google Cloud, AWS, and Azure.



Multi-cloud by practice, not by slide

Imran works and teaches across all three major clouds — Google Cloud, AWS, and Azure — and holds instructor credentials on each side of the fence: **Google Cloud Authorized Trainer** and **Google Certified Professional Data Engineer**, **AWS Authorized Instructor (AAI)**, and **Microsoft Certified Trainer (MCT)**. That matters in this room: your world is SQL Server, SSIS, and Power BI, and your target is BigQuery — he is fluent in both, and can translate between them instead of reading you a Google script.

Large-scale data engineering — the craft

- **Production pipelines at national scale.** He architects and builds production data pipelines over high-volume, multi-source datasets — distributed processing with PySpark and cloud data platforms feeding mission-critical, high-stakes workflows where reliability and auditability are not optional.
- **A PhD in data processing.** His doctoral research attacked the core problem of moving and processing very large datasets across distributed computing infrastructure — culminating in **a large-scale data-processing algorithm of his own design, built on cloud/grid computing.**
- **Pioneered techniques for large-scale processing.** That work introduced scheduling and data-movement techniques for processable bulk data that were ahead of their time — published in peer-reviewed venues (ACM, IEEE) and applied to real data-intensive science, including experiments generating **neutrino-particle data.**
- **A trainer of data teams.** Hundreds of corporate engagements worldwide — including data engineering and cloud training for **Citibank, JP Morgan, the United Nations, Deloitte, and the European Commission** — most of them, like yours, senior teams modernizing from on-prem platforms to cloud data stacks.

AI and agents — the second act

- **Author of four Packt books**, including *30 Agents Every AI Engineer Must Build*, the bestselling *50 Algorithms Every Programmer Should Know*, and *Architecting AI Software Systems*.
- **Builds agentic AI systems for high-stakes, mission-critical workflows** — LLM-based extraction and triage systems delivered to production validation teams.

- **ATLAS** — an open-source, production-grade Agentic RAG platform (LangGraph, corrective RAG, self-verifying answers), built as a public showcase of industrial LLM engineering standards.
- Authored a 9-module *Designing and Deploying AI Agents* curriculum with a full lab series, and an *Agentic Security: Attack and Defend AI Agents* course; 32K+ followers read his writing on applied AI.

In the classroom

Visiting Professor at the University of Ottawa, and a long-tenured authorized instructor with ROI Training. His calibration rule for this room: *you are senior in data engineering and new to one console — so we skip re-teaching what you own and spend our time on where things live in GCP.*

"I build large data systems for a living. These two days map that craft onto Google Cloud — through your migration, not a generic tour."

The books



30 Agents Every AI Engineer Must Build (Packt) — thirty production-ready agent architectures: perception, memory, reasoning, planning, multi-agent collaboration, and ethical alignment. The full source code and labs are open: github.com/PacktPublishing/30-Agents-Every-AI-Engineer-Must-Build. His other titles: *50 Algorithms Every Programmer Should Know* (the bestseller, second edition), *40 Algorithms Every Programmer Should Know*, and *Architecting AI Software Systems*.

Everything from class — links worth keeping

Link	What it is
survey.anum.cloud	Pre-course survey — fill in your row before Monday 9:00 AM
github.anum.cloud	This student pack — schedules, notes, lab companions, slides (Word + PDF)
meet.google.com/ogk-ivdv-bhi	Google Meet — join here both days, 9:00 AM CT
Google Skills classroom	All hands-on labs (temporary GCP

Link	What it is
	projects)
Google check-in	Google's attendance record for the class
On-demand credits	50 free lab credits per training day, valid 30 days — re-run the labs after class
Course evaluation	3 minutes at the end of Day 2
Book code	All 30 agents, open source

Google Cloud & data engineering — the reading list I hand to data teams

Curated from the ROI instructor community's *Google Cloud Helpful Links* doc, filtered to what your migration actually touches.

BigQuery, end to end

- [BigQuery introduction](#) and [BigQuery under the hood](#) — start here
- [Standard SQL reference](#) — your T-SQL habits, translated
- [Partitioned tables](#) and [clustered tables](#) — your tuning toolkit
- [Performance best practices](#), [query plan explanation](#), and [controlling costs](#)
- [Batch loading](#), the [Data Transfer Service](#), and [scheduled queries](#)
- [External data on Cloud Storage](#) and [slots, explained](#)
- [bigquery-utils](#) — community UDFs worth stealing, and the [BigQuery sketchnote](#) for your wall

Migration and replication

- [Datastream overview](#), [supported sources](#), and the [replicate-to-BigQuery quickstart](#)
- [Storage Transfer Service](#) and [on-premises transfers](#) — for the historical bulk moves

Dataform — your Silver→Gold tool

- [Dataform overview](#), [SQL workflows](#), [assertions](#), and the [release lifecycle](#) that maps to your CI/CD discipline

Cloud Storage — your Bronze layer

- [Introduction](#), [best practices](#), [lifecycle management](#), and the [gcloud storage CLI](#)

Staying current

- [Google Cloud blog](#) — data analytics, [release notes](#), and [what's new every week](#)
- [ROI Training's Google Cloud certification roadmap](#) — where to go after these two days (the 3-day Data Warehousing course is next)

- [Looker Studio with BigQuery](#) — for the Power BI authors curious about the Google-side BI option

The same services, on video — five-minute official intros

Every link is from Google's own channels — Google Cloud Tech and Google Cloud — and most run about five minutes; the "in a minute" ones run sixty to ninety seconds. Start with the services your migration actually touches.

BigQuery, end to end

- [What is BigQuery? \(2:27\)](#) or the 86-second [BigQuery in a minute](#) — start here
- [The BigQuery API landscape \(7:33\)](#) — includes the Data Transfer Service chapter at 5:17
- [BigLake in a minute \(2:02\)](#) — the lakehouse bridge to Day 2
- [BigQuery ML in a minute \(1:41\)](#) and [Vertex AI + BigQuery ML \(3:39\)](#) — models in plain SQL
- [The bq command-line tool \(5:17\)](#) — your sqlcmd habit, translated; [Cloud Shell Editor \(2:44\)](#) — where you'll run it

Migration and replication

- [Introduction to Datastream for BigQuery \(5:40\)](#) — CDC, the extract-killer
- [MySQL → BigQuery with Datastream \(2:11\)](#) — the two-minute setup demo
- [What is Storage Transfer Service? \(3:24\)](#) and [Getting started \(4:56\)](#) — for the historical bulk moves

Dataform — your Silver→Gold tool

- [Streamline Your SQL Workflows with Dataform \(7:00\)](#) — repositories, workspaces, release configs

Cloud Storage — your Bronze layer

- [Bucket options \(3:27\)](#) — naming, locations, storage classes
- [How to write data into Cloud Storage \(3:09\)](#) — every upload path
- [How to store data on Google Cloud \(6:52\)](#) — where GCS fits next to Cloud SQL, Firestore, BigQuery, Spanner, and Bigtable

Streaming and pipelines

- [Cloud Pub/Sub Overview \(6:28\)](#) or the [fundamentals segment \(5:09\)](#) — topics and subscriptions
- [What is Dataflow? \(5:03\)](#) — the Drawing Board episode; [Dataflow for Real-time ETL \(5:29\)](#) — the 2024 follow-up

Orchestration and events

- [Moving Cron Jobs to the Cloud \(3:20\)](#) — Cloud Scheduler ≈ SQL Agent

- [Introducing Cloud Workflows \(4:37\)](#) — serverless YAML orchestration
- [Cloud Composer \(3:59\)](#) and [Composer for data orchestration \(3:59\)](#) — managed Airflow
- [Eventarc: build an event-driven orchestration](#) and [Creating Eventarc Triggers](#) — events without plumbing

Databases on the Google side

- [Meet Cloud SQL](#) — managed PostgreSQL, MySQL, and SQL Server
- [AlloyDB Omni quickstart \(5:33\)](#) and [Is AlloyDB compatible with PostgreSQL? \(2:30\)](#) — the OLTP sibling
- [What is Cloud Spanner? \(6:18\)](#) — global relational, when regional isn't enough
- [What is Cloud Bigtable? \(5:37\)](#) — wide-column NoSQL; [the 2-minute version](#)
- [Get started with Firestore \(12:39\)](#) — document NoSQL; [the 2-minute version](#)

The wider landscape — named in class, not drilled

- [Data science workloads on Dataproc Serverless \(6:28\)](#) — the optional Spark lab's engine; [Dataproc in a minute \(1:27\)](#)
- [Data Fusion in a minute \(1:44\)](#) and [Dataprep in a minute \(1:39\)](#) — code-free pipelines and data wrangling
- [Exploring PaaS with App Engine \(7:25\)](#) — or [the 72-second version](#)

Governance and operations

- [De-identifying data with Sensitive Data Protection \(4:39\)](#) — the modern DLP; [DLP in a minute \(2:05\)](#)
- [What is IAM Policy? \(4:36\)](#) — permissions, Allow vs Deny; [the 10-minute IAM deep dive](#)
- [Getting started with Cloud Monitoring \(4:18\)](#) and [Creating alerts \(5:42\)](#)

Deeper cuts — when the five minutes aren't enough

- [Manage complex SQL workflows in BigQuery \(20:41\)](#) — the Dataform long talk
- [Database configurations with Cloud SQL \(17:00\)](#) — decision tree through tuning
- [Enterprise data integration with Cloud Data Fusion \(22:53\)](#) — the no-code session
- [Run Spark and Hadoop faster with Dataproc \(16:35\)](#)
- [Cloud Tasks and Cloud Scheduler, getting started \(13:17\)](#) — Scheduler content starts at 5:38

Scan to connect



[linkedin.com/in/cloudanum](https://www.linkedin.com/in/cloudanum) — course follow-ups welcome. That's where the agents writing, the book updates, and the weekly applied-AI newsletter live.